

Digital Lock-In Amplifiers

OE1022D-DSP Lock-In Amplifier



Features

- **2 independent input channels**
- **2 signal generators**
- **1 mHz to 102 kHz frequency range**
- **1 nV to 1 V full-scale sensitivity**
- **Time constants from 10 μ s to 3 ks**
- **>100 dB dynamic reserve**
- **Automatic adjustment**
- **Multiple-harmonic measurement**

Overview

OE1022D DSP Lock-in Amplifier provides a superb performance within its bandwidth from 1 mHz to 102 kHz. With the advantage of the latest digital signal processing technology and high-precision 24-bit ADC, OE1022D can easily detect the phase and the magnitude of weak signals overwhelmed by various large noise. The performance of OE1022D is as good as other lock-in amplifiers all over the world, even better than them in some certain parameters, such as measurement accuracy, SNR, dynamic reserve. More importantly, the OE1022D has two independent input channels and two independent high-precision signal generators. Each input channel and signal generator can be used independently, which is equivalent to a traditional lock-in amplifier. This means that the OE1022D is equivalent to two traditional lock-in amplifiers. However, thanks to the twin symmetrical design, the two independent input channels and signal generators have ultra high synchronicity, which meets the measurement requirements demanding extremely high synchronization. This performance is not achievable in two traditional lock-in amplifiers.

Input Channel

Two independent input channels have high synchronicity and can be individually configured as a single-ended mode or a differential voltage mode. With an ultra low-noise pre-amplifier, the input noise is as low as 6 nV/ $\sqrt{\text{Hz}}$ @997 Hz. The input impedance is 10 M Ω and the full-scale input voltage sensitivity ranges from 1 nV to 1 V. Besides, OE1022D can be used for current measurement with gains of 10⁶ or 10⁸ V/A. Two line filters (50/60 Hz and 100/120 Hz) are designed to eliminate power frequency interference. A programmable gain amplifier is used to adjust the dynamic reserve, so that OE1022D can keep a high dynamic reserve of 100 dB. The high-precision 24-bit ADC has a sampling rate of 312.5 kSPS, and the excellent anti-aliasing filter in front of the ADC can effectively prevent signal aliasing.

Reference Channel

Two independent reference channels can work in external mode or internal mode. In internal mode, a precise and stable internal oscillator generates sine wave as an internal reference that is multiplied by the input signal. This internal signal is without any phase noise. With the digital phase-shifting technique, the phase resolution of the reference signal is 0.01°. OE1022D can work at any fixed frequency from 1 mHz to 102 kHz in this mode. In external mode, the reference signal can be a sine wave or a TTL pulse or a square wave. The rising or falling edge of the external reference signal triggers the Phase Lock Loop (PLL) to lock the external signal. Based on the frequency of the reference signal, the OE1022D can detect the harmonics of the input signal. The maximum harmonic signal frequency can reach 32767 times the fundamental frequency, and the maximum harmonic frequency cannot exceed the maximum operating frequency of the instrument by 102 kHz. In addition, the OE1022D has a single-channel reference mode, in which two independent input channels are locked and measured using the same external reference channel (REF IN A). This mode can further meet the need for higher synchronization requirements.

Digital Demodulator and Output Filter

The key component of the OE1022D is the digital demodulator. Compared to traditional analog lock-in amplifiers, the OE1022D's internal digital demodulator effectively rejects the measurement errors caused by DC drift and offset. In addition, by optimizing the multiplication of the internal coherent signal of the digital demodulator, the calculation error is minimized so that the instrument can accurately detect the input weak signal. Time constants of the output low-pass filter from 10 μ s to 3 ks can be selected with a choice of 6, 12, 18 or 24dB/oct rolloff. This low-pass digital filter is implemented using an high performance digital filter with a sample rate of 312.5 kHz. The digital demodulation and the low-pass filter used in OE1022D guarantees a high

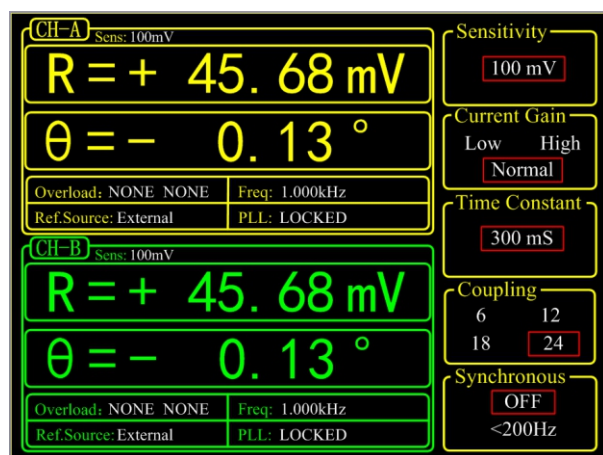
Digital Lock-In Amplifiers

OE1022D-DSP Lock-In Amplifier

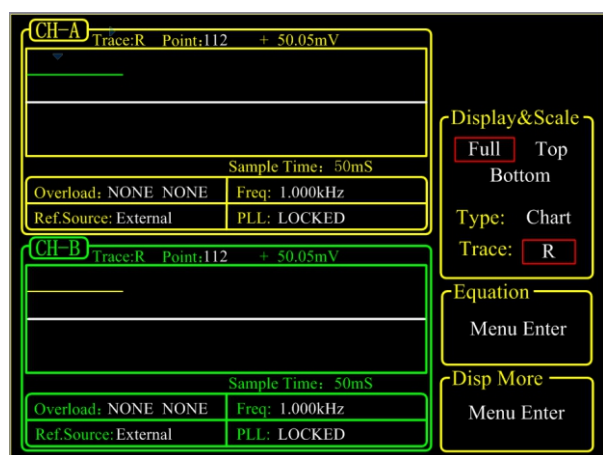
dynamic reserve ($>100\text{dB}$), accurate phase (absolute phase error $<1^\circ$). Moreover, when the frequency of the input signal is lower than 200 Hz , A synchronous filter can be used to eliminate the harmonic influence of the reference signal, ensuring that OE1022D can detect a low-frequency signal quickly and effectively.

Display

OE1022D has a 5.6-inch 640 x 480 color TFT-LCD. The measurement results of OE1022D, such as X, Y, R, and θ , are shown in numerical form, bar graph, X-Y chart on the display.



In X-Y chart, OE1022D shows the trend of measurement results over time, and check the value by using knob control cursor.



Internal Oscillator

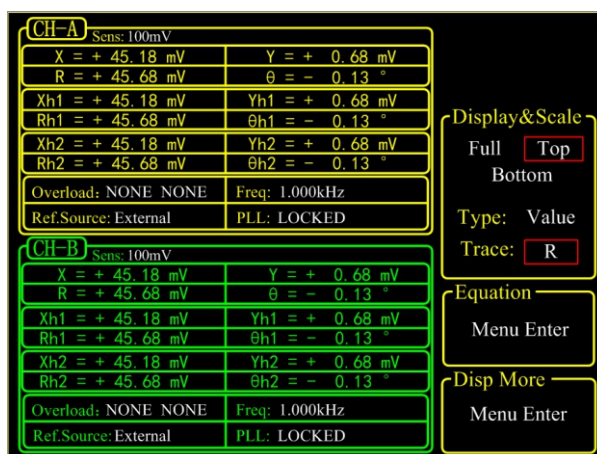
The internal oscillator of OE1022D generates a low distortion (-80 dBc) sine reference signal varying from 1 mHz to 102 kHz , which has a high frequency resolution of 1 mHz . The frequency and amplitude of the reference signal can be set by using the front panel of OE1022D or communication interface. When OE1022D is set in the external reference mode, the internal reference signal is phase-locked with the external reference signal.

Signal Generator

OE1022D uses two high precision digital-to-analog converters (DACs) to output two sine wave signals at the same frequency as their corresponding internal oscillators. The amplitude and phase of the output sine wave can be set through the OE1022D's display, where the maximum amplitude of the sine wave is 5 Vrms .

Simultaneous Multiple-harmonic Measurement

In the traditional lock-in amplifiers, only the fundamental frequency signal or a certain harmonic signal can be measured at one time, so it can not meet the requirement of multiple-harmonic measurement in some occasions. On the contrary, OE1022D uses a flexible digital framework combined FPGA and ARM, which make it practicable and efficient to measure 3 harmonic components simultaneously for each input channel, which means that each input channel is equivalent to three traditional lock-in amplifiers. Because of two independent input channels in OE1022D, OE1022D can detect 6 harmonics (2 fundamentals and 4 harmonics) at one time. The maximum harmonic signal frequency can reach 32,767 times the fundamental frequency, but the maximum harmonic frequency cannot exceed the maximum operating frequency of the instrument by 102 kHz .



Manual Operation

The parameters are convenient to be adjusted by the soft keys besides the display and the numeric keypad on the front panel, such as the internal oscillator frequency and the SINE OUT amplitude.

Auto Function

OE1022D can automatically adjust itself into different optimal operating modes for different input signals, such as Auto Gain mode, Auto Reserve mode and Auto Phase mode. This function makes it easier for users to measure signals more efficiently.

Digital Lock-In Amplifiers

OE1022D-DSP Lock-In Amplifier

Interface

OE1022D uses RS-232 and USB 2.0 as standard interfaces. Through communication interfaces, all instrument functions can be controlled and all data can be read in real-time. Meanwhile, all interfaces of OE1022D are distributed on the front panel and the rear panel.

Remote Operation

Users can use PC to control OE1022D through communication interfaces, including setting the parameters and reading the measurement data. OE1022D is equipped with a free LabVIEW program, which makes it easy to use in complex scientific experiments.

OE1022D LabVIEW interface showing configuration and data display sections.

Configuration Section:

- Input/Filter Configuration:** Includes input source (Single Voltage Signal, Floating, AC Coupled, Input Filter), dynamic range (Normal, 100 mV/nA), sensitivity (100 mV/nA), and filter settings (Filter Time Constant: 300 ms, Filter Slope: 12 dB/oct, Filter Order: 1).
- Reference Configuration:** Includes reference signal source, reference frequency, and reference phase.
- Output/Sampling Configuration:** Includes output mode, sampling rate, and output resolution.
- Formula Configuration:** Includes formula selection and calculation mode.

Data Display Section:

- Ch-A:** X (Vrms) 0.000, Y (Vrms) 0.000, R (Vrms) 0.000, θ (deg) 0.00, Freq (Hz) 0, Noise (Vrms) 0.000.
- Ch-B:** X (Vrms) 0.000, Y (Vrms) 0.000, R (Vrms) 0.000, θ (deg) 0.00, Freq (Hz) 0, Noise (Vrms) 0.000.
- AUX-ADC:** AUX-ADC1 (V) 0.000, AUX-ADC2 (V) 0.000, AUX-ADC3 (V) 0.000, AUX-ADC4 (V) 0.000.
- Data Sampling Rate:** 0.1 S.

OE1022D Specifications

Two Signal Channels

Voltage input Mode	Single-ended or Differential
Full-scale Sensitivity	1 nV to 1 V in a 1-2-5 sequence 1 fA to 1 μ A
Current input	10^6 or 10^8 V/A
Impedance	
Voltage	10 M Ω // 25 pF, AC or DC coupled
Current	1 k Ω to virtual ground
C.M.R.R	>100 dB to 10 kHz, decreasing by 6 dB/oct
Dynamic reserve	>100 dB
Gain accuracy	0.2% typ, 1% max
Noise	6 nV/ $\sqrt{\text{Hz}}$ at 997 kHz 13 fA/ $\sqrt{\text{Hz}}$ at 997 Hz 15 fA/ $\sqrt{\text{Hz}}$ at 97 Hz
Line filters	50/60 Hz and 100/120 Hz
Grounding	BNC shield can be grounded or floated via 10 k Ω to ground

Two Reference Channels

Input	
Frequency range	1 mHz to 102 kHz
Reference input	TTL or Sine
Input impedance	1 M Ω /25 pF
Phase	
Resolution	0.01°
Absolute phase error	<1°
Relative phase error	<0.01°
Orthogonality	90° $\pm$ 0.001°
Phase noise	
Internal ref.	Synthesized, <0.0001°rms at 1 kHz
External ref.	0.001°rms at 1 kHz (100 ms time constant, 12 dB/oct)
Drift	<0.01°/°C below 10 kHz <0.1°/°C above 10 kHz
Harmonic detection	2F, 3F, ...nF to 102 kHz (n<32767)
Acquisition time	
Internal Ref.	Instantaneous acquisition
External Ref.	(2 cycles + 5 ms) or 40 ms, whichever is larger

Demodulator

Stability	
Digital outputs	no zero drift on all setting
Display	no zero drift on all setting
Analog outputs	<5 ppm/°C for all dynamic reserve settings
Harmonic rejection	-90 dB

Time constants	10 μ s to 3 ks (<200 Hz) 10 μ s to 30 s (>200 Hz) (6, 12, 18, 24 dB/oct rolloff)
Synchronous filters	Available below 200 Hz (18, 24 dB/oct rolloff)

Internal Oscillator

Frequency	
Range	1 mHz to 102 kHz
Accuracy	2 ppm + 10 μ Hz
Resolution	1 mHz
Distortion	-80 dBc (f<10 kHz), -70 dBc (f>10 kHz)
Amplitude	0.001 to 5 Vrms
Accuracy	1%
Stability	50 ppm/°C
Outputs	Sine output on rear panel TTL sync output on rear panel

Display

Screen	5.6 inch, 640 $\times$ 480 TFT
Screen format	Single or dual display
Display quantities	Each display shows one trace, traces can be defined as X,Y,R, θ
Display types	Numerical form, bar graph and strip chart

Two Outputs

CH1 and CH2 Outputs	
Function	X, Y, R, θ
Output Voltage	$\pm$ 10 V full scale. 30 mA max output current

Interfaces

USB2.0 and RS232 interfaces

General

Power requirements	
Voltage	220/240 V AC 100/120 V AC(optional)
Frequency	50/60 Hz
Power	50W
Dimensions	
Width	473 mm
Depth	490 mm
Height	
With feet	160 mm
Without feet	147 mm
Weight	11 kg